

Analytic continuation of eigenvalue problems

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In this paper we consider the dependence of Schrödinger equation eigenvalue problems on the coupling-constant parameters in the potential. We show that unless great care is taken, analytic continuation in these parameters can lead to surprising and paradoxical conclusions. Careful analysis of the analytical properties shows that such eigenvalue problems can have elaborate internal structure; these problems can incorporate several different eigenvalue problems joined together. For example, the anharmonic oscillator, whose potential is $V(x) = a^2x^6 - 3ax^2$, is actually four different eigenvalue problems combined into one.

An apparent paradox may arise in the analytic continuation in the coupling-constant parameters of some quantum-mechanical potentials. As a simple example, consider the potential

$$V(x) = a^2x^6 - 3ax^2. \quad (1)$$

The Schrödinger equation eigenvalue problem corresponding to this potential has the form

$$-\psi''(x) + V(x)\psi(x) = E\psi(x). \quad (2)$$

For this potential, when $a > 0$, the ground-state wave function ψ_0 is known *exactly*:

$$\psi_0(x) = \exp(-\frac{1}{4}ax^4). \quad (3)$$

(This eigenfunction corresponds to the lowest-lying energy eigenvalue because it has no nodes.) The ground-state energy $E_0(a)$ is exactly 0. Now suppose we analytically continue the simple function

$$E_0(a) = 0 \quad (4)$$

from positive to negative values of the parameter a ; apparently, the result is still $E_0(a) = 0$. However, this conclusion is clearly wrong! It is easy to see that the spectrum of the potential $V(x) = a^2x^6 - 3ax^2$ for $a < 0$

is strictly positive. Indeed, the lowest eigenvalue E_0 of this potential can be found numerically:

$$E_0 = c|a|^{1/2}, \quad (5)$$

where $c = 1.9355\dots$ ^{#1, #2}. How can it be that the analytic continuation of the zero function is not zero?

To answer this question we must define very carefully the procedure of analytic continuation of a Schrödinger equation eigenvalue problem. Clearly, if we continue the wave function $\psi_0 = \exp(-\frac{1}{4}ax^4)$ to a negative value of a we obtain a nonnormalizable

^{#1} A simple variational calculation for the ground-state energy of the potential $V(x) = a^2x^6 + 3ax^2$, $a > 0$, using the Gaussian trial wave function $\psi(x) = \exp(-Bx^2)$ gives the approximation $e_0 = (\sqrt{2a/5}\sqrt{3})(4 + \sqrt{6})\sqrt{1 + \sqrt{6}} = (1.956\dots)\sqrt{a}$. A better trial wave function is $\psi(x) = \exp(-Bx^2 - \frac{1}{4}ax^4)$; a calculation using this function as a zeroth approximation in a nonlinearization procedure (see ref. [5]) gives $E_0 = (1.9355\dots)\sqrt{a}$.

^{#2} It is important to state here that when $a > 0$ the potential $V(x) = a^2x^6 - 3ax^2$ has the form of a double well. However, the ground-state wave function $\psi_0(x) = \exp(-\frac{1}{4}ax^4)$ is sharply peaked and has a *single* maximum at $x = 0$. The ground-state wave function has a similar form when $a < 0$. For this case the potential is a simple single well.

solution to the Schrödinger equation, which indicates that a simple replacement of a by $-a$ is too naive.

The correct procedure for the analytic continuation of a Schrödinger equation eigenvalue problem was first discussed by Bender and Wu [1] for the case of the anharmonic oscillator potential^{#3} $V(x) = x^2 + gx^4$. Using the procedure of ref. [1] we argue as follows: Consider the more general potential^{#4}

$$V(x) = a^2 x^6 + 2abx^4 + (b^2 - 3a)x^2. \quad (6a)$$

(In the opening discussion we considered the special simple case where $b=0$.) The Schrödinger equation eigenvalue problem for this real potential has the form of a differential equation

$$-\psi''(x) + V(x)\psi(x) = E(a, b)\psi(x) \quad (6b)$$

and an associated boundary condition imposed on the real axis

$$\lim_{|x| \rightarrow \infty} \psi(x) = 0. \quad (6c)$$

In preparation for continuing the parameter a into the complex plane we must generalize the eigenvalue problem from the real- x axis to the complex- x plane. To do so we examine the possible asymptotic behaviors of $\psi(x)$ for large $|x|$. From (6a) and (6b) we see that as $x \rightarrow \infty$ the two possible asymptotic behaviors of the wave function $\psi(x)$ are $\psi_+(x) \approx \exp(\frac{1}{4}ax^4)$ and $\psi_-(x) \approx \exp(-\frac{1}{4}ax^4)$. Thus, when $a > 0$ the eigenfunctions of (6) behave like $\exp(-\frac{1}{4}ax^4)$ and satisfy the asymptotic condition in the complex- x plane

$$\begin{aligned} \psi_-(x) \rightarrow 0 \quad \text{as } |x| \rightarrow \infty \quad \text{if } |\arg x| < \frac{1}{8}\pi \\ \text{and } \frac{7}{8}\pi < \arg x < \frac{9}{8}\pi. \end{aligned} \quad (7)$$

(To be precise, note that if $b > 0$, we also have $\psi_-(x) \rightarrow 0$ as $|x| \rightarrow \infty$ when $\arg x = \pm \frac{1}{8}\pi, \frac{7}{8}\pi, \frac{9}{8}\pi$.)

Thus, we have extended the eigenvalue problem in (6) from the real axis to two wedge-shaped regions in the complex- x plane. These regions appear on fig.

1 as shaded areas. Note also the surprising result that if the asymptotic behavior of the wave function $\psi(x)$ is given by $\psi_-(x) \approx \exp(-\frac{1}{4}ax^4)$ then $\psi_-(x) \rightarrow 0$ as $|x| \rightarrow \infty$ with $\frac{3}{8}\pi < |\arg x| < \frac{5}{8}\pi$. Thus, there is *another* eigenvalue problem in a pair of wedge-shaped regions that contain the imaginary- x axis. These additional regions in the complex- x plane are also indicated on fig. 1.

The differential equation (6b) is second order and therefore $\psi(x)$ has *two* possible asymptotic behaviors for large $|x|$. The other possible asymptotic behavior, $\psi_+(x) \approx \exp(\frac{1}{4}ax^4)$ (remember that $a > 0$), satisfies $\psi_+(x) \rightarrow \infty$ as $|x| \rightarrow \infty$ in the shaded regions on fig. 1. However, in the unshaded regions on fig. 1 $\psi_+(x) \rightarrow 0$ as $|x| \rightarrow \infty$. Thus, if we extend the differential equation in (6) into the complex plane we actually obtain four completely independent eigenvalue problems centered about the four lines (i) $\text{Im } x = 0$, (ii) $\text{Re } x = 0$, (iii) $\text{Im } x = \text{Re } x$, and (iv) $\text{Im } x = -\text{Re } x$. In the first two problems $\psi_-(x) \rightarrow 0$ as $|x| \rightarrow \infty$ in the shaded regions and in the second two problems $\psi_+(x) \rightarrow 0$ as $|x| \rightarrow 0$ in the unshaded regions.

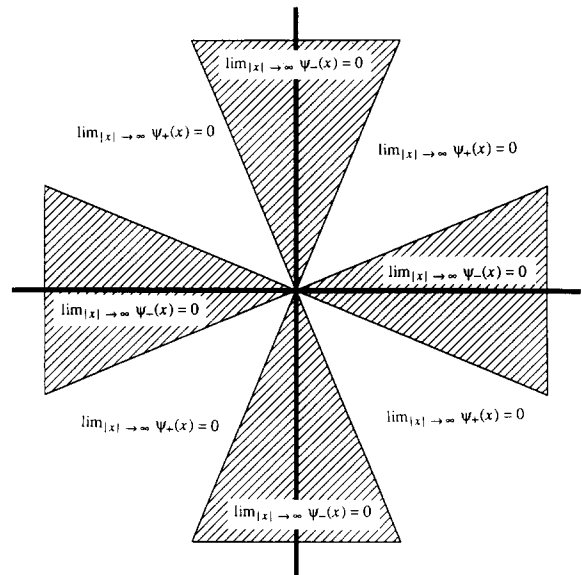


Fig. 1. Regions (shaded) in the complex- x plane where the boundary condition $\psi(x) \rightarrow 0$ is satisfied as $|x| \rightarrow \infty$ for the eigenvalue problem in (6). In this case the parameter a in the potential $V(x)$ given in (6a) is positive and the asymptotic behavior of the wave function $\psi(x)$ for large $|x|$ is $\exp(-\frac{1}{4}ax^4)$. Note that the real and imaginary axes are included in these regions.

^{#3} For this potential a rigorous mathematical discussion of the analytic continuation was given by Loeffel and Martin [2]. See also ref. [3].

^{#4} This potential is the simplest member of a large family of potentials called quasi-exactly-soluble potentials. See ref. [4].

What are the solutions to these four problems? In case (i) corresponding to the line $\text{Im } x=0$, we can solve the differential equation in (6) exactly for the lowest eigenvalue:

$$\psi_{(i)}(x) = \exp\left(-\frac{1}{4}ax^4 - \frac{1}{2}bx^2\right) \quad (8a)$$

and

$$E_{(i)}(a, b) = b. \quad (8b)$$

We know that $E_{(i)}(a, b) = b$ is the lowest eigenvalue because *on the real axis* in the complex- x plane the wave function $\psi_{-}(x)$ in (7a) has no nodes. The other eigenvalues form an unbounded discrete set extending to ∞ .

Next we consider the eigenvalue problem (ii) along the line $\text{Re } x=0$. Here we must consider the differential equation in (6a) and (6b) in the shaded region of the complex- x plane containing the imaginary axis. We perform a rotation to the real axis by letting $x=ir$, where r is a *real* variable with $-\infty < r < \infty$. The new differential equation is

$$\left(-\frac{d^2}{dr^2} + a^2r^6 - 2abr^4 + (b^2 - 3a)r^2 + E\right)\psi(x) = 0, \quad (9a)$$

subject to the boundary condition that

$$\psi(r) \rightarrow 0 \quad \text{as } |r| \rightarrow \infty. \quad (9b)$$

Now observe that the eigenvalue problem in (9) is the same as the eigenvalue problem in (6) except that b has been replaced with $-b$ and that E has been replaced with $-E$. Hence the eigenspectrum for this problem is negative.

Along the line $\text{Im } x = \text{Re } x$ (case (iii)) we let $x = r \exp(\frac{1}{4}\pi i)$ to obtain the eigenvalue problem

$$\left(-\frac{d^2}{dr^2} + a^2r^6 - 2a\beta r^4 + (\beta^2 + 3a)r^2 - iE\right)\psi(x) = 0, \quad (10)$$

where $\beta = ib$ and r is a real variable with $-\infty < r < \infty$. Note that (10) is the same as the eigenvalue problem in (6) except that E has been replaced by $-iE$ and a has been replaced by $-a$. This problem can be solved numerically, if necessary, using variational methods (see, for example, ref. [5]). Along the line

$\text{Im } x = -\text{Re } x$ (case (iv)) we obtain the complex conjugate of case (iii).

Now, finally, we discuss the analytic continuation of the eigenvalue problems in cases (i)–(iv) as the parameter a is rotated into the complex plane. If we let

$$a = \rho \exp(i\theta) \quad (11)$$

and allow θ to increase from 0 to π , then the center line of the region in fig. 1 along which $|\psi(x)| \rightarrow 0$ as $x \rightarrow \infty$ also rotates in the complex plane but in the opposite (clockwise) direction. For example, consider the problem in case (i). If we rotate a by an angle π then from the form of the eigenfunction $\psi_{(i)}(x)$ in (8a) we see that the line along which $\psi(x) \rightarrow 0$ has rotated into the complex plane; it is now the line $\text{Re } x = -\text{Im } x$. Thus, the rotated eigenvalue problem (i) has the form

$$\left(-\frac{d^2}{dx^2} + a^2x^6 - 2abx^4 + (b^2 + 3a)x^2 - E\right)\psi(x) = 0, \quad (12)$$

with $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$, $x = r \exp(-\frac{1}{4}\pi i)$, r real.

To solve this problem we let $x = r \exp(-\frac{1}{4}\pi i)$ in (12) and we obtain

$$\left(-\frac{d^2}{dr^2} + a^2r^6 - 2iabr^4 + (-b^2 - 3a)r^2 + iE\right)\psi(r) = 0. \quad (13)$$

Note that this is the *same* as the problem in case (i) except that b has been replaced by $-ib$ and E has been replaced by $-iE$. Now, if we wish to solve the problem in (13) we can simply use the solution in (8b); to wit, $-iE = -ib$, or $E = b$. As expected, E does not change as $a \rightarrow -a$ because E is *not* a function of a .

However, simply replacing a by $-a$ in (6) and ignoring the rotation of a into the complex plane gives a new eigenvalue problem:

$$\left(-\frac{d^2}{dx^2} + a^2x^6 - 2abx^4 + (b^2 + 3a)x^2 - E\right)\psi(x) = 0. \quad (14)$$

This is no longer the eigenvalue problem in case (i); rather it is the eigenvalue problem in case (iii) rotated clockwise by the angle $\frac{1}{4}\pi$ onto the x -axis. The

ground-state energy for this eigenvalue problem is *not* b as in case (i). Rather, it must be determined numerically. This is the resolution of the paradox raised at the beginning of this paper. Merely replacing a by $-a$ in the differential equation eigenvalue problem converts the original eigenvalue problem into an entirely new and different eigenvalue problem not related by analytic continuation to the one in case (i).

The harmonic oscillator provides a very simple illustration of this phenomenon. Consider the quadratic eigenvalue problem

$$\left(-\frac{d^2}{dx^2} + \frac{1}{4}a^2x^2 - E\right)\psi(x) = 0, \quad (15a)$$

with the associated boundary condition imposed on the real axis

$$\lim_{|x| \rightarrow \infty} \psi(x) = 0. \quad (15b)$$

The ground-state eigenfunction is $\psi(x) = \exp(-\frac{1}{4}x^2a)$ and the ground-state energy is $E = \frac{1}{2}a$.

Now suppose that we replace a by $-a$. The differential equation eigenvalue problem (15) apparently remains the same but the eigenvalue $\frac{1}{2}a$ changes sign! How can this happen?

The answer again is that one may not simply replace the parameter a by its negative in the differential equation and expect to obtain the correct analytic continuation. Instead, one must examine the asymptotic behavior of $\psi(x)$ to determine the region in the complex- x plane in which $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$. This region is shown as the shaded area in fig. 2. We now set

$$a = \rho \exp(i\theta)$$

and rotate θ from 0 to π . As $\arg a$ increases from 0 to π the center line of the region in which $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$ rotates in a clockwise direction. When θ reaches π , the original eigenvalue problem, posed in the shaded area in fig. 2, is now posed in the *unshaded* area:

$$\left(-\frac{d^2}{dx^2} + \frac{1}{4}a^2x^2 - E\right)\psi(x) = 0, \quad (16a)$$

$$\lim_{x \rightarrow \pm i\infty} \psi(x) = 0. \quad (16b)$$

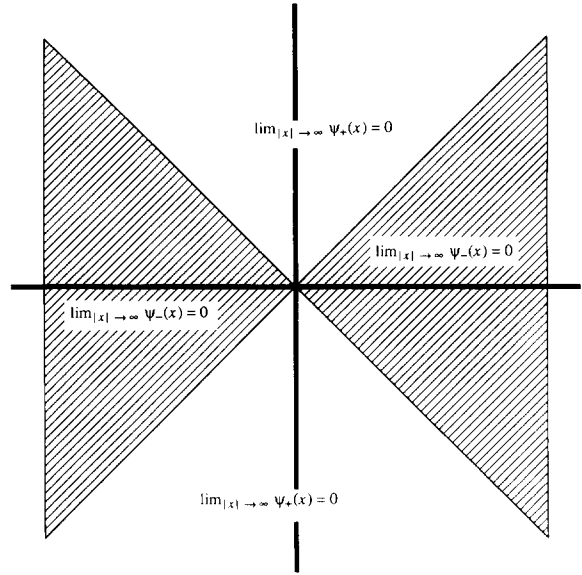


Fig. 2. Regions (shaded) in the complex- x plane where the boundary condition $\psi(x) \rightarrow 0$ is satisfied as $|x| \rightarrow \infty$ for the eigenvalue problem in (15). Note that the harmonic oscillator is actually *two* problems, one on the real axis and the other along the imaginary axis.

To solve this problem we let $x = ir$ and obtain from (16)

$$\left(-\frac{d^2}{dr^2} + \frac{1}{4}a^2r^2 + E\right)\psi(r) = 0, \quad (17a)$$

$$\lim_{r \rightarrow \pm \infty} \psi(r) = 0. \quad (17b)$$

Notice that (17) is the same as (15) except that the sign of E is reversed. The ground-state energy of (17) is

$$E = -a, \quad (18)$$

as it should be. The problem obtained by replacing a by $-a$ in (15) is actually the independent eigenvalue problem posed in the unshaded area of fig. 2.

There is an obvious generalization of the results discussed in this paper. For a polynomial potential $V(x)$ whose highest-power-in- x term is λx^{2n} , there are $n+1$ independent eigenvalue problems in the complex- x plane. As λ rotates to an angle θ in a counterclockwise direction into the complex plane, the array of eigenvalue problems rotates in a clockwise fashion by an angle of $\theta/(2n+2)$. All $n+1$ eigen-

value problems must pass through the real- x axis before the original eigenvalue problem is restored at $\theta = 2(n+1)\pi$ ^{#5}.

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References

- [1] C.M. Bender and T.T. Wu, Phys. Rev. 184 (1969) 1231.
- [2] J.J. Loeffel and A. Martin, in: CERN report no. CERN-TH-1167 (1971), unpublished.
- [3] J.J. Loeffel, A. Martin, B. Simon and A. Wightman, Phys. Lett. B 30 (1969) 656;
B. Simon, Ann. Phys. (NY) 58 (1970) 79.
- [4] A. Turbiner, Commun. Math. Phys. 118 (1988) 467.
- [5] A. Turbiner, Usp. Fiz. Nauk 144 (1984) 35.

^{#5} This fact is a consequence of the Symanzik scaling law discussed in refs. [1–3].